CrossCheck

Cited claim verification over Apollo and Space Race Wikipedia articles

REFERENCE ARCHITECTURE

Databricks Free Edition | serverless notebook

Live golden set: 6/6 correct | 100% citation coverage

01 PREPARE A SMALL, REUSABLE CORPUS

WikipediaLoader

8 topic queries, one string per call
5 results/query; 40 article global cap
Deduplicate by article URL

JSON corpus snapshot

Article text + title + source URL
Content hash and retrieval timestamp
Reuse snapshot for demo and evaluation

Passages + embeddings

1,000 characters; 150 overlap
DatabricksEmbeddings
databricks-bge-large-en

FAISS dense index

In-memory index of short passages
Reuse within the notebook session
Keep source metadata attached

retrieval reads the same index

02 LANGGRAPH WORKFLOW

AUTHOR_PROMPTS (extractor + verifier)

Assertions

User-written paragraph
Claims to check
Example: mission dates

Claim extractor

ChatDatabricks
Atomic claims + queries
AUTHOR_PROMPTS extractor

Retriever

Claim + entity query
Rank fusion; 4 passages
Max 2 per source

Verifier + checks

ChatDatabricks + Python
AUTHOR_PROMPTS verifier
Exact quotes; provenance

Quality metrics

Citation coverage
Abstentions and errors
Potential conflicts

Cited report

Deterministic HTML
Claim / verdict / quote
Links from source data

if useful

One bounded retry

Only insufficient evidence + a new query
Search FAISS; append up to 2 passages
Reverify, then finish regardless of verdict

finish

State

claims | evidence | verdicts | query history
retry count | metrics | trace | report

Input questions need explicit assertions first.
At most 6 claims; at most one follow-up per claim.

Live golden-set run

Databricks Free Edition, serverless compute
LLM: databricks-meta-llama-3-1-8b-instruct
Embeddings: databricks-bge-large-en
6/6 golden claims correct; citation coverage 100%

Prompt engineering loop (offline, promptfoo)

Business goal + labelled evals drive the prompt text
30 labelled verifier claims (supported / refuted / insufficient)
8 extractor paragraphs with reference atomic claims
Same 8B Databricks endpoint and notebook message format
5 scored tuning rounds; a regressing variant was dropped
Final verifier 26/30 (87%); final extractor 8/8
Each round targets an observed failure mode
Winning prompts are pasted into AUTHOR_PROMPTS

03 DIAGNOSTIC EVALUATION AND INTERPRETATION

Hand-labelled cases

Six candidate claims; labels supplied by author
Enter the same graph at retrieval (skip extractor)
Score accuracy and per-class results
Inspect extraction separately on the demo

What the checks establish

Exact quotes must occur in a retrieved passage.
Supported/refuted results require cited evidence.
Invalid output earns no evaluation credit.
Quote presence does not establish entailment.

How to read the results

Coverage is enforced; also inspect abstention rate.
No decided claims means coverage is N/A.
Conflicting stances flag potential disagreement.
Shared Wikipedia sources are not independence.

Two vector queries = multi-query dense retrieval. Evaluation uses the frozen corpus; it is not run recursively inside the application.